# (In)sensitivity to Surface-Level Heuristics:
## *A Case from Turkish Verbal Attractors*


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Preprint

### Same suffix, opposite outcomes

Identical -lAr on nominal vs. verbal attractors yields attraction only for nominals.

### Nominal status matters, not just form

Whether -lAr marks a noun — not phonological similarity — determines interference. Controller-association drives retrieval.

### Cross-linguistic convergence

Controller-association unifies Russian, Czech, English, and Turkish: attraction occurs only when overlap coincides with controllerhood.

## Background

**Agreement attraction** — plural-marked nouns interfere with subject–verb agreement, causing errors (Bock & Miller 1991). **But does the surface form of the plural marker matter?**

### Syncretism & Phonological Overlap

Some languages show more attraction when the attractor's form overlaps with the nominative/plural. But the pattern is **mixed:**

- **German & Czech:** Syncretism with nominative boosts attraction (Hartsuiker et al. 2003; Chromý et al. 2023; Lacina et al. 2025; also see Frank et al. 2010; Avetisyan et al. 2020).
- **English pseudoplurals** (*cruise*) do not attract (Bock & Eberhard 1993).
- **Russian:** Sycnretism between GEN.SG and NOM.PL induce attraction (Slioussar 2018) — but GEN.PL (no overlap) does not.
- **Czech GEN.SG:** Same overlap as Russian, but **no** attraction effect (Lacina & Chromý 2022).

### How could phonological similarity play a role in processing?

**Phonological modulation** (Slioussar 2018): Surface overlap co-activates +NOM and +PL features, boosting retrieval interference. But: fails to explain the Czech null.

**Controller-association** (Lago et al. 2019): What matters is whether a form is distributionally linked to *agreement controllerhood* — not phonological overlap. Supported by Hindi (Bhatia & Dillon 2022), Romanian (Bleoţu & Dillon 2024), English (Schlueter et al. 2018).

## Turkish: The Critical Test

- Turkish -lAr is **identical** on nominal and verbal forms. Crucially, both can be standalone subjects, and the verbal -DIK forms pattern nominally in several diagnostics (Demirok 2019) — but only nominal -lAr **controls agreement**. This makes Turkish ideal: it dissociates surface overlap from controller potential with minimal confounds.

**NOMINAL -LAR**

Plural on nouns. Can be a subject **and** controls verb agreement: strong controller-association.

**VERBAL -LAR**

Agreement on nominalized verbs. Can be subject, but **cannot** control agreement: not controller-like.

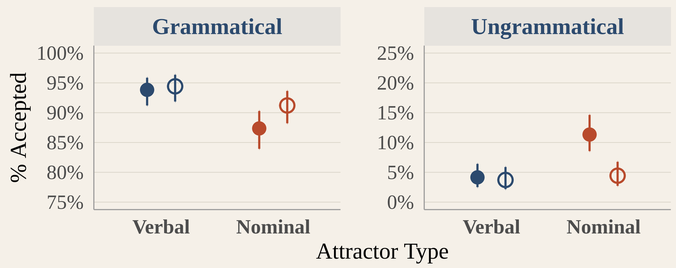
| | ATTRACTOR (PL/SG) | HEAD | … VERB |
|---|---|---|---|
| **Verbal** | Tut-tuk-(**lar**/Ø)-ı | aşçı | … zıpla-dı(-lar) |
| **Nominal** | Milyoner-(**ler**/Ø)-in | aşçı-sı | … zıpla-dı(-lar) |

TRANSLATION

*'The cook that (s/he|they) hired jumped(*-PL)…' / 'The millionaire('s|s') cook jumped(*-PL)…'*

## Results: Experiment 1

**Exp 1: N** = 80, 40 items, RSVP (400 ms/word), speeded acceptability judgments; 2×2×2 design.
**Exp 2:** verbal-only replication, **N** = 95, same task, testing the verbal null without nominal items.




**NOMINAL**

Plural attractor nearly **triples** false acceptances: 4% → 11%. Classic attraction.

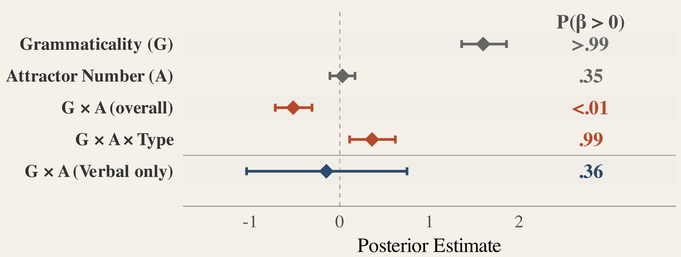
**VERBAL**

**No effect:** 4% vs 4%. Identical -lAr suffix produces zero interference.

**DISSOCIATION**

Same suffix, opposite outcomes — surface form is not enough.

Bayesian GLMM — Posterior Estimates




**KEY INTERACTION**

Attraction **differs by type:** nominal shows a 12-point increase; verbal shows none.

**VERBAL NULL**

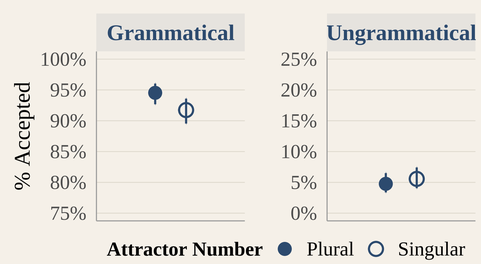
No verbal attraction. P(β>0) = .36 — posterior centered on zero.

**OVERALL**

Attraction confined to nominal attractors only.

## Results: Experiment 2 (Verbal Only)

Overall acceptability was lower than in previous Turkish studies (Lago et al. 2019, Türk 2022, Türk & Logačev 2024), consistent with trial composition shifting response criteria (Hammerly et al. 2019; Arehalli & Wittenberg 2021). This motivated a **verbal-only** replication (**N** = 95).




**REPLICATION**

**No attraction.** G × A: β = −0.01, P(β>0) = .46. Null holds even as the only source.

**STRONGER TEST**

Verbal-only design eliminates any nominal contamination. N = 95.

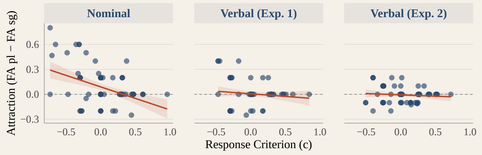
## Quantifying the Verbal Null

$BF_{01}$ comparing models **with ($H_1$)** vs. **without ($H_0$)** G × A. All priors favor the null.

| PRIOR | $BF_{01}$ | INTERPRETATION |
|---|---|---|
| Wide: N(0, 2.0) | 12.40 | **strong for $H_0$** |
| Main: N(0, 1.0) | 4.73 | moderate for $H_0$ |
| Narrow: N(0, 0.25) | 2.73 | anecdotal for $H_0$ |

*Stable across 5 prior specifications ($BF_{01}$ = 2.73–12.40). Replicated in Exp 2 verbal-only design.*

- **Response bias control:** The verbal null is not a criterion artifact — attraction is uncorrelated with individual response bias.




## Key Conclusions

- **Nominal status is what matters:** Identical -lAr produces attraction only on nominal attractors — not verbal ones. Category, not surface form, determines interference.
- **Distribution matters:** Attraction is stronger when a form is linked to the kinds of nominals that serve as agreement controllers.
- **Verbal -lAr is the critical contrast:** Same suffix, but no attraction. That is the central dissociation.

## Cross-Linguistic Evidence

| LANGUAGE | PHON. OVERLAP | CONTROLLER | ATTRACTION? |
|---|---|---|---|
| Russian (Gen.SG) | Yes | Yes | **Yes** |
| Czech (Gen.SG) | Yes | No | **No** |
| English (*cruise*) | Yes | No | **No** |
| Turkish (Verbal) | Yes | No | **No** |
| Turkish (Nominal) | Yes | Yes | **Yes** |

**Pattern: Attraction occurs when phonological overlap coincides with controller-association.**

## Theoretical Accounts

**Phonological Modulation (Slioussar 2018)**
Surface overlap should boost attraction for both nominal and verbal -lAr. Turkish suggests this account needs amendment.

**Marking & Morphing (Eberhard et al. 2005)**
Verbal -lAr is agreement morphology rather than an inherent plural controller, so a verbal null is expected. Harder to extend to syncretism effects cross-linguistically.

**Noisy-Channel / Rational Inference**
Also compatible with the verbal null, but less specific about why nominals and verbals with the same suffix diverge.

**Cue-Based Retrieval + Controller-Association** ✓
Best fit: attraction tracks forms associated with nominal agreement controllers. Explains why nominal -lAr attracts but verbal -lAr does not.